Literature Review

2.1 Introduction to HR Analytics

Human Resource Analytics (HR Analytics) refers to the process of collecting, organizing, analyzing, and interpreting employee-related data to support better human resource management decisions.

Organizations increasingly use HR analytics to understand workforce trends, evaluate employee performance, measure satisfaction, and identify potential challenges affecting employee retention.

According to research by Margherita (2022), HR analytics covers different areas, including descriptive, diagnostic, and predictive analysis, helping organizations understand workforce-related information and its potential business value.

In this project, HR analytics is used to examine employee data through Microsoft Power BI, transforming raw HR records into meaningful metrics, visualizations, and insights that support workforce planning and employee retention strategies.

2.2 Employee Attrition & Employee Retention

Employee attrition refers to the reduction in an organization's workforce when employees leave their positions for different reasons, such as resignation, retirement, or other circumstances.

Employee retention, on the other hand, refers to an organization's ability to keep its employees and maintain a stable workforce.

High employee attrition can create several organizational challenges, including increased recruitment and training costs, loss of experience, reduced productivity, and additional workloads for remaining employees.

Research by Bolt et al. (2022), in their systematic review of more than 1,300 studies on labour turnover, highlights the importance of understanding the factors associated with employees leaving organizations. Their review also demonstrates the long-standing role of employee turnover in human resource management research.

Understanding attrition patterns is therefore important for organizations seeking to improve workforce stability and develop effective employee retention strategies.

2.3 Factors Associated with Employee Attrition

Employee attrition is influenced by multiple personal and organizational factors. Understanding these factors helps HR departments identify potential retention challenges.

2.3.1 Job Satisfaction & Work Environment

Job satisfaction represents how employees feel about their jobs and their overall work experience.

Factors such as workplace conditions, relationships with colleagues, management support, and job responsibilities can influence employee satisfaction.

Employees experiencing lower job satisfaction may be more likely to consider leaving their organizations.

Research on employee turnover consistently identifies job satisfaction as an important factor associated with employees' intentions to leave.

This project examines job satisfaction, environment satisfaction, and relationship satisfaction to understand how these factors differ between employees who stay and those who leave.

2.3.2 Compensation & Working Conditions

Compensation is another important aspect of employee retention. Employees may evaluate their salaries and financial benefits in relation to their responsibilities, experience, and perceived contribution to the organization.

Working conditions, including overtime, business travel, and distance from home, may also influence employees' overall work experience.

Differences in compensation and working conditions can be associated with different attrition patterns across departments and job roles.

Therefore, this project investigates salary distribution, overtime, business travel frequency, and distance from home to identify potential relationships with employee attrition.

2.3.3 Work-Life Balance & Overtime

Work-life balance refers to the ability of employees to manage their professional responsibilities while maintaining their personal lives.

Excessive working hours and frequent overtime may create additional pressure and negatively affect employee well-being.

Research on employee turnover intention highlights work-life balance as one of the factors that can influence employees' decisions and intentions regarding their employment.

This project explores work-life balance ratings and overtime patterns to determine whether employees experiencing less favorable working conditions have higher attrition rates.

2.3.4 Performance & Employee Evaluation

Performance evaluation is an important HR practice used to assess employee contributions, identify strengths, and recognize areas requiring improvement.

Performance ratings can also help organizations understand differences in employee development, compensation, and career progression.

Comparing employee self-ratings with manager ratings can provide additional insights into differences in performance perceptions and evaluation practices.

This project examines performance ratings across departments and job roles and investigates whether performance levels are associated with salary increases and employee attrition.

2.3.5 Training & Career Development

Training and career development provide employees with opportunities to improve their skills, enhance their performance, and progress professionally.

Organizations that provide development opportunities may strengthen employees' perceptions of organizational support and encourage long-term commitment.

A meta-analysis by Ng et al. (2024) found that employer-sponsored career development practices are associated with employee performance and turnover outcomes, partly through perceived organizational support and organizational attachment.

Based on this research, the project investigates training frequency, promotion history, and salary increases to explore their relationships with employee performance and retention.

2.4 The Role of Data Analytics in HR Decision-Making

Data analytics enables HR departments to move beyond traditional reporting toward more informed and evidence-based decision-making.

Through descriptive and diagnostic analysis, organizations can:

- Monitor workforce size and employee turnover.
- Identify departments and employee groups with higher attrition rates.
- Compare employee satisfaction and performance levels.
- Examine relationships between compensation, working conditions, and employee retention.
- Identify potential areas for improvement in HR policies.

Research by Belizón (2024) discusses HR analytics as a knowledge discovery process that helps organizations extract useful information from HR data.

In this project, Microsoft Power BI is used to organize HR metrics, visualize workforce patterns, and provide interactive dashboards that allow management to explore employee-related information and make more informed decisions.

2.5 Related Studies & Previous Research

Several academic studies have investigated HR analytics, employee turnover, and the factors associated with employee retention.

Study 1: A Century of Labour Turnover Research: A Systematic Literature Review (2022)

Authors: Bolt et al.

This study reviewed more than 1,300 research papers related to labour turnover. It examined the development of turnover research and the different approaches used to understand why employees leave organizations.

Relevance to this project: It provides a research foundation for examining employee attrition and understanding the importance of analyzing multiple factors rather than relying on a single indicator.

Study 2: Human Resources Analytics: A Systematization of Research Topics and Directions for Future Research (2022)

Author: Margherita.

This study reviewed HR analytics research and organized its major topics into areas such as analytics enablers, applications, and organizational value.

Relevance to this project: It supports the use of data analytics to transform HR information into insights that contribute to better organizational decisions.

Study 3: Employer-Sponsored Career Development Practices and Employee Performance and Turnover: A Meta-Analysis (2024)

Authors: Ng et al.

This research examined the relationship between employer-sponsored career development practices, employee performance, and turnover. The findings indicate that career development support can strengthen organizational attachment and contribute to better employee outcomes.

Relevance to this project: It supports investigating training opportunities, promotions, and salary increases as part of the employee retention analysis.

Study 4: Human Resources Analytics in Practice: A Knowledge Discovery Process (2024)

Author: Belizón.

This study explored how HR analytics is applied in organizations using a knowledge discovery approach. It highlights how HR-related information can be analyzed to generate useful organizational knowledge.

Relevance to this project: It supports the project's analytical workflow, from understanding and preparing employee data to identifying meaningful patterns and insights.

2.6 Research Gap & Project Contribution

Previous research has extensively explored employee turnover and its relationship with different organizational and personal factors. Research has also highlighted the growing importance of HR analytics in supporting evidence-based decision-making.

However, organizations still need practical ways to combine different HR data sources and present workforce insights in a clear and accessible format.

This project addresses this practical need by bringing together employee information related to demographics, attrition, satisfaction, compensation, performance, training, and career development within an interactive Power BI dashboard.

The project contributes by:

- Providing a comprehensive overview of workforce composition and attrition patterns.
- Investigating multiple factors associated with employee attrition in one analytical solution.
- Examining performance, training, and career development alongside employee satisfaction and working conditions.
- Highlighting data quality issues that may affect the reliability of HR analysis.
- Presenting findings through interactive visualizations that support HR decision-making.

The final objective is to provide a practical analytical solution that helps HR management understand workforce trends, identify potential retention challenges, and develop more informed employee retention strategies.

References

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